Jack Richter

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EDUCATION

University of Michigan Ann Arbor, MI
Ph.D. Climate and Space Sciences and Engineering *2022 – present*
M.S. Climate and Space Sciences and Engineering *2025*

University of Wisconsin Madison, WI
B.S. Atmospheric and Oceanic Sciences (AOS) *2022*
Cert. Computer Sciences, Cert. Mathematics *2022*

RESEARCH EXPERIENCE

Graduate Student Research Assistant *University of Michigan* *2022 – present*
Cold Precipitation Processes Research Group

- Advised by Prof. Claire Pettersen
- Investigating links between large-scale atmospheric conditions and snow microphysics using ground-based, airborne, and satellite observations

Undergraduate Researcher *University of Wisconsin* *2021 – 2022*
Space Science and Engineering Center

- Advised by Dr. Claire Pettersen
- Characterized the impacts of inland atmospheric rivers on the macro- and micro-physical properties of snowfall in Marquette, MI

PUBLICATIONS AND DATASETS

Richter, J., Pettersen, C., McMurdie, L., Bailey, A., Shates, J. Ice Microphysics in Stratiform and Convective Precipitation Regions during the IMPACTS Field Campaign, *Journal of Applied Meteorology and Climatology*, (submitted October 2026)

Wagner, T., **Richter, J.**, Mateling, M., Pettersen, C., Kulie, M. (2026). The Synoptic, Microphysical, and Radar Characteristics of Upper Great Lakes Snow Events, *Journal of Geophysical Research-Atmospheres,* https://doi.org/10.1029/2026JD046349.

Richter, J., Pettersen, C., Moisseev, D., Shates, J., King, F. (2025). Macro- and Microphysical Properties of Atmospheric River Snowfall: Results from Two Instrument Sites, *Journal of Applied Meteorology and Climatology*. https://doi.org/10.1175/JAMC-D-24-0220.1.

Richter, J., Pettersen, C. (2024). Marquette, Michigan Snowfall Instrument Site: Micro Rain Radar Dataset [Data set], *University of Michigan - Deep Blue Data*. https://doi.org/10.7302/4qdq-8184.

AWARDS

AMS Student Presentation Honorable Mention *17th Conference on Cloud Physics* *2026*

- Awarded by the AMS Conference on Cloud Physics Committee for the oral presentation *Ice Microphysics in Stratiform and Convective Precipitation Regions during the IMPACTS Field Campaign*

Rackham Conference Travel Grant *University of Michigan* *2023 – 2026*

- Received $4,000 in support across four years for travel to various science conferences

AGU Outstanding Student Presentation Award *2024 Annual Meeting* *2025*

- Awarded by the AGU Hydrology Section for the oral presentation *Investigating Atmospheric River Snowfall Properties Using Suborbital Platforms*

AGU Precipitation Student Award *2024 Annual Meeting* *2025*

- Awarded by the AGU Precipitation Technical Committee for the oral presentation *Investigating Atmospheric River Snowfall Properties Using Suborbital Platforms*

NASA FINESST Fellowship *Earth Sciences Division* *2024*

- Funded to investigate the proposal: *Leveraging Global Precipitation Measurement Mission Assets to Evaluate Atmospheric River Snowfall Properties*

Graduated with Distinction *University of Wisconsin* *2022*

- Top 20% of graduating class

Lettau-Wahl Scholarship Award *University of Wisconsin, AOS.* *2021*

- Award is given for excellent performance as an undergraduate

Fluno Scholar Custer Award *University of Wisconsin, Marching Band* *2020*

- Presented to members of the University of Wisconsin Marching Band who demonstrate leadership and commitment to the success of the ensemble

Johnson Scholarship Award *University of Wisconsin, AOS* *2020*

- Given to an exemplary AOS undergraduate who intends to pursue graduate studies

Dean's List *University of Wisconsin, College of Letters and Science* *2018 – 2022*

- Fall 2018, Spring 2019, Fall 2020, Spring 2021, Fall 2021, Spring 2022

Academic Excellence Scholarship *State of Wisconsin HEAB* *2018*

- Awarded to Wisconsin high school seniors who have the highest grade point average in each public and private high school throughout the State of Wisconsin.

PRESENTATIONS

2026

- **Richter** et al., "Ice Microphysics in Stratiform and Convective Precipitation Regions during the IMPACTS Field Campaign", **AMS 17th Conference on Cloud Physics**, Madison, Wisconsin, 2026, *Oral*

- **Richter** et al., "Macro- and Microphysical Properties of Atmospheric River Snowfall: Results from Two Instrument Sites", **4th (AC)³ Science Conference on Arctic Amplification**, Cologne, Germany, 2026, *Poster*

2025

- **Richter** et al., "Synthesizing Airborne Observations from IMPACTS to Examine Snow Microphysics in Regions of Enhanced Integrated Water Vapor Transport", **University of Michigan Climate Week Poster Session**, Ann Arbor, Michigan, 2025, *Poster*
- **Richter** et al., "Synthesizing Airborne Observations from IMPACTS to Examine Snow Microphysics in Regions of Enhanced Integrated Water Vapor Transport", **AMS 41st International Conference on Radar Meteorology**, Toronto, Canada, 2025, *Poster*
- **Richter**, "Investigating Atmospheric River Snowfall Properties Using Suborbital Platforms", **AGU Precipitation Technical Committee 2nd Quarterly Seminar**, Online, 2025, *Oral* (**invited**)

2024

- **Richter** et al., "Investigating Atmospheric River Snowfall Properties Using Suborbital Platforms", **AGU Annual Meeting**, Washington, D.C., 2024, *Oral*
- **Richter** et al., "Impacts of Atmospheric Rivers on Snow Microphysics", **Michigan Geophysical Union**, Ann Arbor, Michigan, 2024, *Poster*

2023

- Olson et al. (including **Richter, J.**), "Validation of Ice-Phase Precipitation Estimates Derived from GPM Satellite Combined Radar-Radiometer Observations", **AGU Annual Meeting**, San Francisco, California, 2023, *Oral*
- **Richter** et al., "Impacts of Atmospheric Rivers on Snow Microphysics", **Midwest Student Conference on Atmospheric Research**, Urbana-Champaign, Illinois, 2023, *Poster*
- **Richter** et al., "Impacts of Atmospheric Rivers on Snow Microphysics", **AMS 40th International Conference on Radar Meteorology**, Minneapolis, Minnesota, 2023, *Poster*

2022

- **Richter** et al., "Micro- and Macro-physical Properties of Atmospheric River Snowfall Events in the Upper Great Lakes", **AMS 16th Conference on Cloud Physics**, Madison, Wisconsin, 2022, *Poster*

TEACHING AND MENTORING EXPERIENCE

Guest Lecturer *University of Michigan* *2026*

- Course: *Cloud and Precipitation Processes*
- Prepared in class exercise and gave lecture on airborne precipitation observations to a class of undergraduate and graduate students

Grader *University of Michigan* *2023 – 2026*

- Courses: *Earth System Dynamics, Introduction to Atmospheric Radiation,* and *Cloud and Precipitation Processes*
- Graded homework assignments, proctored exams, provided feedback on exam content

Graduate Student Peer Mentor *University of Michigan* *2024*

- Provided new graduate students with peer support during their first year
- Formed positive connections across the department and developed mentoring skills

FIELD EXPERIENCE

S^{2}noCliME *Steamboat Springs, CO* *2025*

- Assisted with various activities during the campaign: preparing/launching radiosondes, instrument/site maintenance, weather forecasting

Snowfall Site Work *Gaylord, MI* *2023*

- Performed routine instrument maintenance on two ground-based instruments: the Precipitation Imaging Package and the Micro Rain Radar

SERVICE AND OUTREACH

PhD Retreat Organizing Committee *2026*

Volunteer Member

- Planned logistics, transportation, and team building activities for the 7th annual Climate and Space PhD Retreat

Graduate Undergraduate Student Organization (GUStO) *2024 – 2026*

Student Engagement Officer

- Organized and led activities related to, but not limited to, student engagement and social events within the Department of Climate and Space Sciences and Engineering
- Coordinated the 14th annual snow pool: collected $1,261 in donations for the Shelter Association of Washtenaw County

Climate Bits and Bites *2024*

Guest Presenter

- Gave a talk on snow microphysics to a group of climate scientists with diverse research backgrounds and career stages

RELEVANT SKILLS

Programming

- Languages: Python (primary), Matlab, Java, GEMPAK
- Systems: Linux, Unix, MacOS

Instruments and Reanalysis

- Ground Based: Precipitation Imaging Package, Micro Rain Radar, Radiometer Physics GmbH radar, Weather Surveillance Radar 1988 Doppler
- NASA ER-2: High-Altitude Wind and Rain Airborne Profiler, Cloud Radar System, ER-2 X-band Doppler Radar
- NASA P-3: High Volume Precipitation Spectrometer, Cloud Droplet Probe, 2D-S Stereo Probe, King Probe, Water Isotope Spectrometer for Precipitation and Entrainment Research

- NASA Global Precipitation Measurement Mission: Dual-frequency Precipitation Radar
- Reanalysis: ERA5, MERRA-2

Radiosondes
- InterMet: iMet-4 and iMet-3050A sounding platform

Weather Forecasting
- Synoptic and mesoscale forecasting of snowfall systems

WORKSHOPS AND SPECIAL TRAININGS

$(AC)^3$ IRTG – PhD Winter school *Hyytiälä, Finland* *2025*
- Causes, contributions, and trends to Arctic Amplification

IMPACTS Data Workshop *virtual* *2023*
- Instrument updates and statuses, ground instruments, aircraft

AMS Radar Short Course *Minneapolis, MN* *2023*
- Open Radar Short Course

Michigan Institute for Data Science *Ann Arbor, MI* *2023*
- Environmental Data Science Summer Academy

Stony Brook University *Stony Brook, NY* *2023*
- Summer School in MM Wavelength Radar Observations

AMS Short Course *virtual* *2022*
- Machine Learning in Python for Environmental Science Problems Short Course

AMS Short Course *virtual* *2022*
- Learning to Engage with the DOE Atmospheric Radiation Measurement User Facility

PROFESSIONAL MEMBERSHIPS

American Geophysical Union *2024 – present*
American Meteorological Society *2021 – present*